

Brave Lab[端點取證]Volatility3、HxD

從一台疑似受感染的 Windows 工作站取得了記憶體映像。該系統屬於一名被標記為潛在惡意活動（包括未經授權的存取嘗試和異常瀏覽模式）的使用者。安全團隊觀察到與加密通訊服務相關的外部 IP 的網路活動。

您的任務是分析提供的記憶體轉儲以發現有關所涉及進程的詳細信息，識別受到攻擊時的活動連接，並追蹤特定應用程式的使用模式。

Tools:

Volatility 3

CertUtil

HxD

起始收集

```
volatility_2.6_win64_standalone.exe -f
C:\Users\user\Downloads\temp_extract_dir\c49-AfricanFalls2\20210430-
Win10Home-20H2-64bit-memdump.mem imageinfo
```

顯示:

```
Volatility Foundation Volatility Framework 2.6
INFO      : volatility.debug      : Determining profile based on KDBG search...
           Suggested Profile(s) : No suggestion (Instantiated with
Win2008SP1x86)
           AS Layer1 : IA32PagedMemoryPae (Kernel AS)
           AS Layer2 : FileAddressSpace
(C:\Users\user\Downloads\temp_extract_dir\c49-AfricanFalls2\20210430-
Win10Home-20H2-64bit-memdump.mem)
           PAE type : PAE
           DTB : 0x3L
KUSER_SHARED_DATA : 0xffdf0000L
```

看起來失敗，要用版本3

按照此頁面安裝:<https://letsdefend.io/blog/how-to-install-volatility-2-and-volatility-3-on-windows>

根據嫌疑系統，RAM 影像是在什麼時間取得的？

指令:python vol.py -f C:\Users\user\Downloads\temp_extract_dir\c49-AfricanFalls2\20210430-
Win10Home-20H2-64bit-memdump.mem windows.info.Info

```
C:\Users\user\Desktop\鑑識工具\volatility3-2.26.0>python vol.py -f C:\Users\user\Downloads\temp_extract_dir\c49-AfricanFalls2\20210430-Win10Home-20H2-64bit-memdump.mem windows.info.Info
Volatility 3 Framework 2.26.0
Progress: 100.00 PDB scanning finished
Variable Value
Kernel Base 0xf8043cc00000
DTB 0x1aa000
Symbols file:///C:/Users/user/Desktop/%E9%91%91%E8%AD%98%E5%B7%A5%E5%85%B7/volatility3-2.26.0/volatility3/symbols/windows/ntkrnlmp.pdb/769C521E4833ECF72E21F02BF33691A5-1.json.xz
Is64Bit True
IsPAE False
layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf8043d80f368
Major/Minor 15.19041
MachineType 34404
KeNumberProcessors 4
SystemTime 2021-04-30 17:52:19+00:00
NtSystemRoot C:\Windows
NtProductType NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Tue Oct 11 07:04:26 1977
```

2021-04-30 17:52 <= 這應該是正確答案，但不曉得一直說錯誤...

第二季

RAM 影像的 SHA256 雜湊值是多少？

```
(root@kali)-[/home/kali/Downloads]
# sha256sum 20210430-Win10Home-20H2-64bit-memdump.mem
9db01b1e7b19a3b2113bfb65e860fffd7a1630bdf2b18613d206ebf2aa0ea172 20210430-Win10Home-20H2-64bit-memdump.mem
9db01b1e7b19a3b2113bfb65e860fffd7a1630bdf2b18613d206ebf2aa0ea172
```

第三季

brave.exe的進程 ID 是什麼？

指令:python vol.py -f C:\Users\user\Downloads\temp_extract_dir\c49-AfricanFalls2\20210430-Win10Home-20H2-64bit-memdump.mem windows.pslist.PsList

```
4856 1872 brave.exe 0xbf0f6ca782c0 0 - 1 False 2021-04-30 17:48:45.000000 UTC 2021-04-30 17:50:56.000000 UTC Disabled
4856
```

第四季

收購時已建立多少個網路連線？

指令:python3 vol.py -f 20210430-Win10Home-20H2-64bit-memdump.mem windows.netscan.NetScan |grep -i "established"

```
(root@kali)-[/home/kali/Desktop/tool/volatility3]
python3 vol.py -f 20210430-Win10Home-20H2-64bit-memdump.mem windows.netscan.NetScan |grep -i "established"
0xbf0f6a53ca20 TCPv4 10.0.2.15DB scan49833fin52.230.222.68 443 ESTABLISHED 2812 svchost.exe 2021-04-30 17:50:07.000000 UTC
0xbf0f6ad16050 TCPv4 10.0.2.15 49829 142.250.191.208 443 ESTABLISHED 5624 svchost.exe 2021-04-30 17:49:58.000000 UTC
0xbf0f6ad1fad0 TCPv4 10.0.2.15 49847 52.230.222.68 443 ESTABLISHED 2812 svchost.exe 2021-04-30 17:52:17.000000 UTC
0xbf0f6c6352b0 TCPv4 10.0.2.15 49842 52.113.196.254 443 ESTABLISHED 5104 SearchApp.exe 2021-04-30 17:51:25.000000 UTC
0xbf0f6c7104d0 TCPv4 10.0.2.15 49778 185.70.41.130 443 ESTABLISHED 1840 chrome.exe 2021-04-30 17:45:00.000000 UTC
0xbf0f6cd4fa20 TCPv4 10.0.2.15 49837 204.79.197.200 443 ESTABLISHED 5104 SearchApp.exe 2021-04-30 17:51:18.000000 UTC
0xbf0f6d0c64a0 TCPv4 10.0.2.15 49843 204.79.197.222 443 ESTABLISHED 5104 SearchApp.exe 2021-04-30 17:51:26.000000 UTC
0xbf0f6d51c4a0 TCPv4 10.0.2.15 49838 13.107.3.254 443 ESTABLISHED 5104 SearchApp.exe 2021-04-30 17:51:23.000000 UTC
0xbf0f6d525a20 TCPv4 10.0.2.15 49845 23.101.202.202 443 ESTABLISHED 1156 MsMpEng.exe 2021-04-30 17:51:36.000000 UTC
0xe80000193a20 TCPv4 10.0.2.15 49845 23.101.202.202 443 ESTABLISHED 1156 MsMpEng.exe 2021-04-30 17:51:36.000000 UTC
```

問5

Chrome 與哪個網域建立了網路連線？

同上，先找到google的連線

0xbf0f6c7104d0	TCPv4	10.0.2.15	49778	185.70.41.130	443
ESTABLISHED	1840	chrome.exe	2021-04-30 17:45:00.000000	UTC	

DNS解析

```
(root@kali)-[/home/kali/Desktop/tool/volatility3]
# nslookup 185.70.41.130
130.41.70.185.in-addr.arpa      name = 185-70-41-130.protonmail.ch.
protonmail.ch
```

問6

PID 6988的程序執行檔的 MD5 哈希值是多少？

指令：python3 vol.py -f 20210430-Win10Home-20H2-64bit-memdump.mem windows.pslist.PsList --pid 6988 --dump

```
(root@kali)-[/home/kali/Desktop/tool/volatility3]
# python3 vol.py -f 20210430-Win10Home-20H2-64bit-memdump.mem windows.pslist.PsList --pid 6988 --dump
Volatility 3 Framework 2.26.2
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output
6988 4352 OneDrive.exe 0xbf0f6d4262c0 26 - 1 True 2021-04-30 17:40:01.000000 UTC N/A 6988.OneDrive.exe.0x1c0000.dmp
(root@kali)-[/home/kali/Desktop/tool/volatility3]
# ls
20210430-Win10Home-20H2-64bit-memdump.mem API_CHANGES.md CODING_STYLE.md doc MANIFEST.in README.md volatility3 volshell.py vol.spec
6988.OneDrive.exe.0x1c0000.dmp CITATION.cff development LICENSE.txt pyproject.toml test vol.py volshell.spec
```

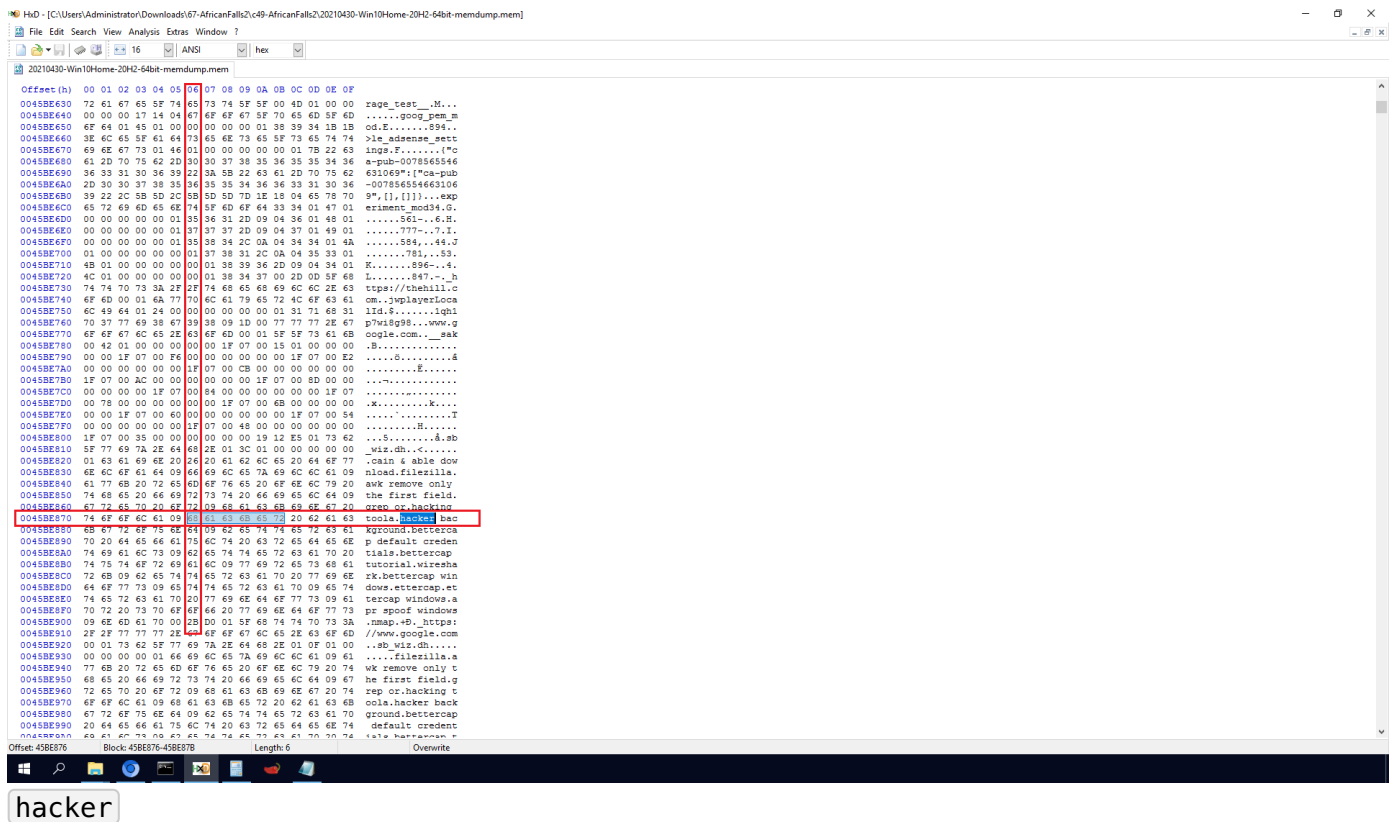
編碼

```
(root@kali)-[/home/kali/Desktop/tool/volatility3]
# md5sum 6988.OneDrive.exe.0x1c0000.dmp
0b493d8e26f03ccd2060e0be85f430af 6988.OneDrive.exe.0x1c0000.dmp
0b493d8e26f03ccd2060e0be85f430af
```

問7

您能辨識從偏移量0x45BE876開始且長度為 6 個位元組的單字嗎？

使HxD



問8

powershell.exe的父行程的建立日期和時間是什麼？

指令：python3 vol.py -f 20210430-Win10Home-20H2-64bit-memdump.mem windows.pslist.PsList

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId
Wow64	CreateTime	ExitTime	File output			
這是子進程						
5096	4352	powershell.exe	0xbf0f6d97f2c0	12	-	1
False	2021-04-30 17:51:19.000000 UTC	N/A	Disabled			
父進程						
4352	4296	explorer.exe	0xbf0f6ca662c0	82	-	1
False	2021-04-30 17:39:48.000000 UTC	N/A	Disabled			

正常來講應該是要寫父才對，但答案父子都是錯的？又是bug

2021-04-30 17:39

問9

記事本中開啟的最後一個檔案的完整路徑和名稱是什麼？

指令：python3 vol.py -f 20210430-Win10Home-20H2-64bit-memdump.mem windows.cmdline

```
2520 notepad.exe "C:\Windows\system32\notepad.exe" C:\Users\JOHND0~1\AppData\Local\Temp\7z04FB31F24\accountNum
1328 chrome.exe "C:\Program Files\Google\Chrome\Application\chrome.exe"
C:\Users\JOHND0~1\AppData\Local\Temp\7z04FB31F24\accountNum
```

Q10

嫌疑人使用Brave瀏覽器多久？（以小時為單位）

指令：python3 vol.py -f 20210430-Win10Home-20H2-64bit-memdump.mem windows.registry.userassist | grep -i Brave

```
python3 vol.py -f 20210430-Win10Home-20H2-64bit-memdump.mem windows.registry.userassist | grep -i Brave
WARNING volatility3.plugins.windows.registry.userassist: UserAssist key not found in \\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT at 0xa80332915000
WARNING volatility3.plugins.windows.registry.userassist: UserAssist key not found in \\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT at 0xa80332aab000
* 0xa80333cda000 \\C:\Users\John Doe\ntuser.dat ntuser.dat\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count 2021-04-30 17:52:18.000000 UTC Value %ProgramFiles%\BraveSoftware\Temp\GUM20E0.tmp\BraveUpdate.exe N/A 0 0 0
:00:03.531000 N/A
* 0xa80333cda000 \\C:\Users\John Doe\ntuser.dat ntuser.dat\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count 2021-04-30 17:52:18.000000 UTC Value %ProgramFiles%\BraveSoftware\Update\BraveUpdate.exe N/A 0 1 0:00:24.797000 N/A
* 0xa80333cda000 \\C:\Users\John Doe\ntuser.dat ntuser.dat\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count 2021-04-30 17:52:18.000000 UTC Value Brave N/A 9 22 4:01:54.328000 2021-04-30 17:48:45.000000 UTC
* 0xa80333cda000 \\C:\Users\John Doe\ntuser.dat ntuser.dat\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{F4E57C4B-2036-45F0-A9AB-443BCFE33D9F}\Count 2021-04-30 17:51:18.000000 UTC Value C:\Users\Public\Desktop\Brave.lnk N/A 8 0 0:00:00.508000 2021-04-30 17:48:45.000000 UTC
```